Week 5
Report assignment

Read the following paper and make a summary and discuss the ethical issues on this BBI device from the points listed today on one page of an A4 size sheet:

Rao, R. P., Stocco, A., Bryan, M., Sarma, D., Youngquist, T. M., Wu, J., & Prat, C. S. (2014). A direct brain-to-brain interface in humans. PloS one, 9(11), e111332.

https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

Summary: Researchers have developed a non-invasive BBI that allows two people to communicate directly with their brains. The interface uses electroencephalography (EEG) to record brain signals from one person (the sender) and transcranial magne c smula on (TMS) to deliver those signals to another person (the receiver). This allows the sender to cause the receiver to perform a motor ac on. In a series of experiments, the researchers found that the brain-to-brain interface was able to transmit informa on with high accuracy. The subjects were able to use the interface to cooperate on a visuomotor task, which they had to work together to achieve a common goal in a computer game. The researchers believe that this is the first direct BBI in humans. Their results provide evidence that it is possible to transmit informa on directly from one brain to another without using any other form of communica on. Ethical Issues: -BBI takes brain input directly from the brain and could result in inappropriate ac ons which would normally be considered but not be executed -The fact the sender can control the receiver's ac ons must be considered if the sender has ill inten ons -Device failure -This is a noninvasive method, however risks s ll can occur